SoccerGMR Performance Improvements: E9 & E9+GEQ vs. GMR Baseline

Direct progressive comparison of SI-Bind (E9 / CAB) and SI-Bind (E9) + GEQ against baseline temporal DETR models on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries).

HIGHLIGHT 1 **SI-Bind (E9 / CAB) vs. Baseline: Feature Drift & Multi-GT Resolution**

- Standard Test: mAP +3.82 pp (13.94% → 17.76%), AUROC +1.97 pp (74.17% → 76.14%)
- Full-Decodable: mAP +4.19 pp (12.23% → 16.42%) across 8 DETR backbones
- Mechanism: Serial CAB Triad (Controlled Refresh → Re-binding → Innovation) stops feature drift.

HIGHLIGHT 2 **SI-Bind (E9) + GEQ vs. Baseline: Dual-Stage Synergy (Ours Full)**

- Standard Test: Orc-Gap -5.65 pp (21.11% → 15.46%), mAP +4.05 pp, Rej-F1 +3.39 pp
- Full-Decodable: Orc-Gap -5.40 pp (20.10% → 14.70%), Rej-F1 +2.83 pp, AUROC +1.34 pp
- Mechanism: Global Event Query grounds sequence energy, suppressing false positive hallucinations.

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Architecture / Method**	**mAP (%) [↑]**	**Orc-Gap (%) [↓]**	**Rej-F1@.5 (%) [↑]**	**AUROC (%) [↑]**	**mAP (%) [↑]**	**Orc-Gap (%) [↓]**	**Rej-F1@.5 (%) [↑]**	**AUROC (%) [↑]**
Moment-DETR	1. GMR Baseline	6.68	24.28	67.94	70.20	5.47	23.48	72.08	78.91
	2. GMR + SI-Bind (E9 / CAB)	13.35 +6.67	21.60 -2.68	63.61 -4.33	74.93 +4.73	**12.99** +7.52	19.56 -3.92	66.27 -5.81	79.12 +0.21
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**13.61** +6.93	**17.61** -6.67	**70.14** +2.20	**75.53** +5.33	**12.99** +7.52	**17.08** -6.40	**73.16** +1.08	**81.84** +2.93
EaTR	1. GMR Baseline	16.49	21.51	65.60	76.56	15.19	20.39	67.38	81.59
	2. GMR + SI-Bind (E9 / CAB)	17.17 +0.68	21.58 +0.07	69.33 +3.73	**77.21** +0.65	**15.25** +0.06	19.81 -0.58	72.41 +5.03	**82.55** +0.96
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**17.23** +0.74	**17.80** -3.71	**73.23** +7.63	75.93 -0.63	15.19 0.00	**16.43** -3.96	**75.87** +8.49	82.47 +0.88
QD-DETR	1. GMR Baseline	14.62	20.23	71.48	**77.01**	12.65	19.05	72.49	81.75
	2. GMR + SI-Bind (E9 / CAB)	16.77 +2.15	24.35 +4.12	59.41 -12.07	75.27 -1.74	**14.96** +2.31	22.68 +3.63	61.60 -10.89	81.78 +0.03
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**17.65** +3.03	**16.73** -3.50	**73.03** +1.55	75.47 -1.54	14.85 +2.20	**15.65** -3.40	**76.11** +3.62	**82.98** +1.23
CG-DETR	1. GMR Baseline	18.65	25.66	57.47	76.40	14.67	23.28	63.23	81.81
	2. GMR + SI-Bind (E9 / CAB)	18.76 +0.11	21.55 -4.11	69.27 +11.80	**77.50** +1.10	**17.41** +2.74	20.21 -3.07	70.53 +7.30	81.24 -0.57
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**19.34** +0.69	**16.84** -8.82	**75.02** +17.55	77.32 +0.92	16.91 +2.24	**15.77** -7.51	**77.41** +14.18	**83.57** +1.76
BM-DETR	1. GMR Baseline	4.00	17.59	**75.00**	71.55	3.65	17.62	78.58	79.11
	2. GMR + SI-Bind (E9 / CAB)	10.72 +6.72	23.09 +5.50	70.85 -4.15	72.73 +1.18	9.54 +5.89	21.78 +4.16	75.34 -3.24	**80.12** +1.01
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**11.87** +7.87	**12.07** -5.52	**75.00** 0.00	**73.43** +1.88	**10.91** +7.26	**11.71** -5.91	**78.82** +0.24	80.04 +0.93
BAM-DETR	1. GMR Baseline	23.25	19.82	70.43	78.66	20.62	18.37	71.79	80.99
	2. GMR + SI-Bind (E9 / CAB)	**24.21** +0.96	20.68 +0.86	69.53 -0.90	**78.96** +0.30	**22.17** +1.55	19.15 +0.78	69.12 -2.67	79.03 -1.96
	3. GMR + SI-Bind (E9) + GEQ (Ours)	23.26 +0.01	**16.16** -3.66	**71.76** +1.33	77.07 -1.59	21.43 +0.81	**15.03** -3.34	**74.57** +2.78	**82.86** +1.87
Sim-DETR	1. GMR Baseline	20.73	22.51	74.69	72.97	19.18	21.78	**75.72**	78.87
	2. GMR + SI-Bind (E9 / CAB)	21.34 +0.61	23.16 +0.65	72.12 -2.57	74.70 +1.73	20.46 +1.28	22.30 +0.52	73.46 -2.26	79.47 +0.60
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**22.45** +1.72	**11.71** -10.80	**75.00** +0.31	**74.86** +1.89	**21.05** +1.87	**11.60** -10.18	75.52 -0.20	**81.43** +2.56
UVCOM	1. GMR Baseline	7.12	17.24	**75.00**	70.03	6.40	16.83	**76.80**	**78.98**
	2. GMR + SI-Bind (E9 / CAB)	**19.74** +12.62	17.28 +0.04	72.89 -2.11	**77.84** +7.81	**18.58** +12.18	16.69 -0.14	69.85 -6.95	75.33 -3.65
	3. GMR + SI-Bind (E9) + GEQ (Ours)	18.51 +11.39	**14.78** -2.46	71.55 -3.45	77.77 +7.74	17.37 +10.97	**14.36** -2.47	69.25 -7.55	77.51 -1.47
FlashVTG	1. GMR Baseline	**25.43**	**25.47**	**66.35**	**74.30**	**20.43**	**24.13**	**68.48**	**82.21**
	2. GMR + SI-Bind (E9 / CAB)		*-- pending --*				*-- pending --*		
	3. GMR + SI-Bind (E9) + GEQ (Ours)								
AVERAGE (8 Backbones)	1. GMR Baseline	**13.94**	**21.11**	**69.70**	**74.17**	**12.23**	**20.10**	**72.26**	**80.25**
	2. GMR + SI-Bind (E9 / CAB)	**17.76** +3.82	**21.66** +0.56	**68.38** -1.33	**76.14** +1.97	**16.42** +4.19	**20.27** +0.17	**69.82** -2.44	**79.83** -0.42
	3. GMR + SI-Bind (E9) + GEQ (Ours)	**17.99** +4.05	**15.46** -5.64	**73.09** +3.39	**75.92** +1.75	**16.34** +4.11	**14.70** -5.40	**75.09** +2.83	**81.59** +1.34

Δ Notation & Badges: Pill badges show metric delta relative to GMR Baseline (Δ = Method - Baseline).
- Green badges [e.g. +6.67 or -5.65] indicate statistically favorable gains: mAP/Rej-F1/AUROC increased, or Orc-Gap decreased.
- Red badges indicate regressions relative to baseline. Bold values mark the best performance across methods for each backbone.

Method 1 (GMR Baseline): Official temporal DETR baseline checkpoints with standard bipartite matching.
Method 2 (GMR + SI-Bind [E9 / CAB]): Stage-1 Triad (Controlled Refresh → Temporal Re-binding → Context Innovation). Resolves feature drift & multi-GT confusion.
Method 3 (GMR + SI-Bind [E9] + GEQ [Ours]): Stage-2 Global Event Query calibration on top of E9. Sequence energy anchoring eliminates false positive hallucinations.
Metric Definitions: mAP (%) over IoU thresholds [0.1:0.1:0.9]; Orc-Gap (%) = area between Oracle envelope and G-mIoU@1 threshold curve across $\tau \in [0, 1]$ (lower is better);
Rej-F1@.5 (%) = F1 score for non-present target moment rejection at default 0.5 presence threshold; AUROC (%) = Area under ROC presence curve. FlashVTG pending.